Dohyun (Eric) Kim

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EDUCATION

University of Auckland **Auckland, New Zealand**
Bachelor of Engineering (Honours) in Computer Systems Engineering *Expected Graduation Nov 2027*

EXPERIENCES

Research Assistant Trainee **Auckland, New Zealand**
University of Auckland *Jul 2026 - Oct 2026*

- Trainee researcher analysing the **robot soccer system** at UoA's CARES Lab, covering camera calibration, **vision processing and decision strategy design**, to support its planned migration to a modern framework.
- Investigated **TurtleBot2** navigation algorithms and implemented LiDAR integration and control using **ROS and Python**, building hands-on experience in autonomous systems.

Robotics Instructor **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) *Apr 2026 - Oct 2026*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

SKILLS

Languages & Tools: C/C++, Python, VHDL, MIPS Assembly, MATLAB, Git, Linux, ROS, AWS
Hardware: Altium Designer, LTSpice, Intel Quartus Prime, ModelSim, Cypress PSoC, Atmel AVR, Proteus, AutoCAD

PROJECTS

A* PhotonTracker **Cypress PSoC & PCB-SMT**
Autonomous Pathfinding Robot - Team Project *Oct 2026*

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Smart Energy Monitor **Embedded C & PCB-TH**
Embedded Energy Monitoring Systems - Team Project *Oct 2025*

- Designed a dual channel energy monitoring system, using **signal conditioned ADC sampling on an ATmega328P**, to compute RMS voltage, peak current, and average power in real time.
- Built and validated a double layer PCB, using **Altium Designer** and UART based output, **to meet a ±5% full scale accuracy spec** across a 2.5 to 7.5 VA load range.

AWARD

3rd Place in ECSE Design Competition 2025 **Team Competition**
"Winnie the Bot" - Interactive Companion Robot *Sep 2025*

- Contributed to an AI powered interviewer robot, using **dual ATmega328P Nano** microcontrollers **to drive face tracking, arm movement, and real time voice interaction.**
- **Prototyped and refined the robot's enclosure**, using iterative 3D printing and multiple design revisions, to deliver a compact, durable, and functional physical build.

LEADERSHIPS

Academic Team Executive **Auckland, New Zealand**
KEB (Korean Engineering Body) - Founding Member *Jul 2024 - Oct 2026*

- **Co-founded** an engineering community of **170+** members; contributed to a **17-member** executive team.
- Delivered **tutorial sessions** for junior engineering students and supported academic event planning.

Official Competition Staff **NZ Robotics Olympiad 2026**
IEEE R&A (IEEE Robotics and Automation Society) *Jul 2026*

- **Selected as official Competition Staff** based on prior volunteering & robotics instructor experience at ciLab.
- Led competition operations and **resolved technical issues**, ensuring smooth running matches for all teams.

Full-time Student Volunteer **NZ Robotics Olympiad 2025**
IEEE (Institute of Electrical and Electronics Engineers) *Jul 2025*

- **Volunteered 40+ hours** supporting operations and logistics at NZRO 2025, an IEEE robotics competition.
- Collaborated with organisers **to ensure a smooth event experience** for participants.